MILITARY SPECIFICATION

BARIUM PERCHLORATE, ANHYDROUS, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one analyzed reagent grade of anhydrous barium perchlorate.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572	-	Wood-Preservative; Water-Repellent.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301	-	Chemicals, Dry and Paste, Packaging and Packing of.

MILITARY

MIL-P-15011	-	Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized and Containerized Unit Loads, 40" x 48" Pallet Skids, Runners, or Pallet-Type Base.

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Freight Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N. W., Washington, D. C. 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Appearance. Anhydrous barium perchlorate shall be a white, crystalline powder when tested as specified in 4.2.4.1.

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3.2 Chemical characteristics. Anhydrous barium perchlorate shall conform to the chemical characteristics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay [as $Ba(ClO_4)_2$]	99.0	----	4.2.4.2
Insoluble matter	----	0.010	4.2.4.3
Chloride (Cl)	----	0.005	4.2.4.4
Nitrogen compounds (as N)	----	0.001	4.2.4.5
Alkalies and calcium	----	0.200	4.2.4.6
Heavy metals	----	0.001	4.2.4.7
Iron (Fe)	----	0.001	4.2.4.8

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the anhydrous barium perchlorate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1/4 pound shall be removed from each sample unit packaging container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of unit packaging containers in batch or lot	Number of sample unit packaging containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified.
(b) Container not as specified.
(c) Container closure not as specified.
(d) Container damaged or leaking.
(e) Strapping or fiberboard packing aids incorrect.
(f) Wood preservation incorrect or missing (level A only).
(g) Marking incorrect, missing, or illegible.

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composite specimen of no less than 1/2 pound. The composite specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.2.4.1 Appearance. Visually examine 10 grams (g) of the specimen for color and form.

4.2.4.2 Assay [as $Ba(ClO_4)_2$]. Weigh to the nearest milligram (mg) 0.9 g of the specimen into a 250-milliliter (ml) beaker. Dissolve with 25 ml of water and filter out any insolubles. Heat the filtrate nearly to boiling and add 5 ml of 1 to 10 sulfuric acid dropwise with vigorous swirling. Digest the precipitate formed on a steam bath for at least 1 hour and cool. Filter with Whatman No. 42 filter paper and wash with three 20-ml portions of 1 to 100 sulfuric acid. Transfer the paper containing the precipitate to a tared crucible and dry in a 110° C oven for at least 1 hour. Ignite the precipitate with free access of air until the residue is white, and then cool. Moisten the precipitate with a few drops of 1 to 10 sulfuric acid and ignite at higher temperature to constant weight. Calculate the percent anhydrous barium perchlorate as follows:

$$\text{Percent } Ba(ClO_4)_2 = \frac{144.1A}{W}$$

where: A = Weight of $BaSO_4$ precipitate in grams, and
W = Weight of specimen in grams.

4.2.4.3 Insoluble matter. Weigh to the nearest milligram approximately 10 g of the specimen and dissolve in 100 ml of water. Heat on a steam bath for 1 hour. Filter through a tared filter crucible weighed to the nearest 0.1 mg, wash with water, dry at 105° to 110° C for 1 hour, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A - B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.4 Chloride. Weigh 2.000 g of the specimen and dissolve in 20 ml of water containing 1 ml of concentrated nitric acid. Divide the solution into two equal portions. To one portion, add 1 ml of 2-percent silver nitrate solution, bring to a boil, allow to stand for 10 minutes, cool to room temperature, filter through nitric acid wash filter paper, and retain the filtrate for use as a control. To the second portion of the specimen solution, add 1 ml of 2-percent silver nitrate solution. Any turbidity produced in the second portion shall be no greater than that produced when 0.05 mg of chloride ion is added to the control and the volumes of both solutions are made equal.

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4.2.4.5 Nitrogen compounds. Weigh 2.000 g of the specimen and dissolve in 40 ml of water in a 100-ml distilling flask. Add 30 ml of 10-percent sodium hydroxide solution and 1.00 g of powdered Devarda's metal. Connect the distilling flask to the distillation apparatus and allow to stand for 1 hour. With the condenser drain beneath water, distill off 40 ml into 10 ml of water. Add to the distillate and water 1 ml of 10-percent sodium hydroxide solution and 2 ml of Nessler's solution. The resulting color shall be no darker than that produced when 0.02 mg of nitrogen ion is treated in the same manner.

4.2.4.6 Alkalies and calcium. Weigh to the nearest milligram approximately 2 g of the specimen and dissolve in 75 ml of water. Add 3 ml of concentrated hydrochloric acid and heat to boiling. Add 15 ml of 10-percent sulfuric acid and allow to stand overnight. Dilute to 100 ml with water, mix well and filter. Evaporate on a steam bath 50 ml of the filtrate in a tared evaporating crucible weighed to the nearest 0.1 mg. Evaporate the solution to dryness, ignite carefully under a hood, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent alkalies and calcium as follows:

$$\text{Percent alkalies and calcium} = \frac{200\ (A - B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.7 Heavy metals. Weigh 4.000 g of the specimen and dissolve in 10 ml of hot water. Add 10 ml of concentrated hydrochloric acid and evaporate to dryness on a steam bath. Dissolve the residue in 40 ml of water. Prepare a control by adding 0.02 mg of lead ion to a 10-ml aliquot of the solution and diluting to 30 ml. For the specimen use the remaining 30-ml portion. To each add 1 ml of 1N acetic acid and 10 ml of freshly prepared hydrogen sulfide solution. The color produced in the solution of the specimen shall be no darker than that produced in the control.

4.2.4.8 Iron. Weigh 4.000 g of the specimen and dissolve in 10 ml of hot water. Add 10 ml of concentrated hydrochloric acid and evaporate to dryness on a steam bath. Dissolve the residue in 40 ml of water. Prepare a control by adding 0.02 mg of iron ion to a 10-ml aliquot of the solution and diluting to 30 ml. For the specimen use the remaining 30-ml portion. To each add 1 ml of 1N acetic acid and 10 ml of freshly prepared hydrogen sulfide solution. Make each solution alkaline with 28-percent ammonium hydroxide. Any green color produced shall be no darker than that produced in the control.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One pound ± 1/2 ounce of barium perchlorate shall be packaged in a type I, class 1 glass bottle, with plastic screw cap, in inclosure a as specified in PPP-C-301. The bottle shall also conform to group A, class 1, type e, style 2, grade 1 with closure A and outer seal A of PPP-C-186.

5.1.1.1 Intermediate packaging. Twelve bottles of barium perchlorate, packaged as specified in 5.1.1, shall be intermediately packaged in accordance with PPP-C-301 except that bottles shall be arranged in one tier of three rows of four bottles per row.

5.1.2 Level C. One pound ± 1/2 ounce of barium perchlorate shall be packaged in a container in a manner to assure integrity of the container and maintenance of the specified purity of the contents, and to provide protection against leakage and hazards under known favorable conditions during shipment, handling, and limited tenures of storage.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Two intermediate packages, prepared as specified in 5.1.1.1, shall be packed upright in a nailed wood box, a cleated-plywood box, or a wirebound wood box. The nailed wood box shall conform to class 2, style 4-1/2, for a type 2 load of PPP-B-621. The cleated-plywood box shall conform to type overseas, style I or J, for a type 2 load of PPP-B-601. The wirebound box shall conform to class 3, style optional, for a type 2 load of PPP-B-585. Fiberboard fillers and pads shall be used as directed in PPP-C-301 to assure a tight pack. Each container shall be closed and strapped with galvanized strapping as specified in the appendix to the applicable box specification. Unless specific provision for preservation is made in the appropriate box specification, each box or the finished, unpainted wood parts thereof shall be completely immersed for no less than 3 minutes in a water-repellent wood preservative conforming to composition A of TT-W-572.

5.2.2 Level B. Barium perchlorate shall be packed as specified in 5.2.1 except the box shall conform to a fiberboard box, style RSC, grade V3c of PPP-B-636 and no preservative shall be used. The box shall be closed and reinforced by pressure-sensitive adhesive filament tape as specified in the appendix to PPP-B-636.

5.2.3 Level C. Forty-eight containers of barium perchlorate, packaged as specified in 5.1.2, shall be packed in containers which comply with the Code of Federal Regulations, the Uniform Freight Classification Rules,

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the National Motor Freight Classification Rules, or other carrier rules as applicable to the mode of transportation. Containers shall be capable of being stacked and of supporting superimposed loads during shipment and shall assure safe delivery to destination without damage to contents. Containers shall be acceptable for shipment at the most favorable rate of the applicable regulation provided that all requirements specified herein have been met.

5.3 Marking. In addition to the special and precautionary marking specified herein, unit, intermediate, and shipping containers shall be marked in accordance with Department of Transportation (DOT) regulations and MIL-STD-129. The yellow label for oxidizing materials as specified by DOT regulations shall be required.

5.3.1 Special marking. Each bottle shall be durably and legibly labeled with the following information:

(a) Empirical formula
(b) Formula weight
(c) Manufacturer's lot number
(d) Assay
(e) Analysis or maximum limits of impurities

5.3.2 Precautionary marking. Each bottle, inclosure, and intermediate and shipping container shall be durably and legibly labeled with the following precautionary marking:

WARNING! STRONG OXIDANT

Contact with combustible material may cause fire.
Can be decomposed by concussion, contact with organic matter, or agents readily subject to oxidation. This decomposition may be explosive in character.
Store separately; keep container closed and away from heat.
Do not take internally.
Avoid contact with skin and eyes.

5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 using a pallet conforming to style 1, type I, class 1 of MIL-P-15011. The pallet and other wood storage aids, if used, shall be treated with wood preservative as specified for the pallet in MIL-P-15011.

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6. NOTES

6.1 Intended use. Anhydrous barium perchlorate is intended for use as a dehydrating agent in laboratory analysis and for use as a general laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 4-167, dated 22 December 1952.

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activity:

Army - SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0553

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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